

Class 2 — Alterations in Cell Function & Growth, Part 2 (Cell replication, cancer & genetics)

NRSG 224 Pathophysiology · Fall 2026 · filled from the class study guide

Personal study sheet — not for submission.

Questions keep the guide's order. Brackets show where each answer came from: S = recorded-lecture slide, mm:ss = recording time (1x), sheet p. = your full study sheet. **not covered in lecture** = textbook knowledge, not from her lecture.

Sources: recorded lecture audio, transcribed by machine (S# = slide, mm:ss = time in the recording), the in-class slides, and your full study sheet pp. 5-7.

Page 1 — Cell replication/differentiation → Cancer

Guide: Mutations in regulatory genes → abnormal proliferation + abnormal differentiation. Example of disorder: cancer. Each printed finding has space under it; fill with what it means / what you'd assess.

1.1 → uncontrolled cell growth

- More cells actively cycling; fewer in the resting phase → uncontrolled growth.
[S10, 08:59-12:15]
- Lose cell density-dependent inhibition; do not die within their normal programmed time frame. [S20, 28:54-30:09]
- Increasing tumor cells crowd out normal cells and steal their blood supply.
[S21, 30:09-30:33; sheet p.6]

1.2 → infiltration into surrounding tissues

- Infiltration = growth extending into surrounding normal tissue; malignant tumors lack a containing capsule. [S38, 50:34-51:10]
- Loss of cell-to-cell cohesiveness → cells break away and travel to other body sites: metastasis. [S37, 47:31-50:34]
- Infiltration makes complete surgical removal difficult; remaining cancer cells keep multiplying → recurrence.

[S37, 47:31-50:34; S38, 50:34-51:10]

1.3 → tissue/organ dysfunction

- Abnormal differentiation → cancer cells do not specialize properly or perform normal tissue functions. [S22, 30:33-31:33]
- Normal cells are crowded out while abnormal cells do not function normally → tissue/organ dysfunction. [S21, 30:09-30:33; S22, 30:33-31:33]

1.4 → changes in cell characteristics

- Cells fail to differentiate/specialize properly → different characteristics and impaired normal function. [S22, 30:33-31:33; S23, 31:33-32:12]
- Lose normal cell-to-cell cohesiveness → break away and metastasize. [S37, 47:31-50:34]
- May produce abnormal substances and display abnormal cell-membrane antigens.

[S24, 32:12-32:31; S29, 36:43-37:09]

1.5 abnormal biochemical products (ex: VEGF, ectopic hormone production, toxins, etc.)

- VEGF = vascular endothelial cell growth factor
→ new blood vessels supply oxygen/nutrients, allowing continued tumor growth.

[sheet p.6; S25, 32:31–33:37]

- Ectopic hormones: cancer cells make hormones without normal negative-feedback regulation → excess hormone/endocrine-like disease. [S26, 33:37–35:49]
- Enzymes/toxins damage tissue → sores that do not heal normally. [S27, 35:49–36:32]
- Abnormal antigens such as PSA and CEA can help monitor cancer progression and treatment response; professor describes this role rather than diagnosis/screening.

[S29, 36:43–37:09; S30, 37:09–40:49]

1.6 abnormal antigens

- Abnormal differentiation → abnormal antigens on cancer-cell membranes; these can serve as clinical markers. [S29, 36:43–37:09]
- Immune system normally fights cancer; immune defects/immunosuppression increase cancer risk. [S18, 25:49–27:10; S44, 58:23–61:14]
- Examples of increased risk: AIDS, immunosuppressive drugs, and aging with less efficient immune function. [S18, 25:49–27:10]

1.7 generalized effects

- Cachexia = weight loss; WBC cytokines cause anorexia/loss of appetite → reduced eating

and weight loss. [S33, 42:10–42:53; S34, 42:53–43:48]

- Pain: tumor pressure on nerves, inadequate blood/oxygen, and tissue damage from enzymes/toxins. [sheet p.6; S35, 43:48–45:34]
- Anemia: poor nutrition (iron/B-vitamin deficiency), bleeding from damaged tissue, or tumor erosion into vessels. [S36, 45:34–47:31]
- Bone-marrow involvement impairs RBC production; cancer treatments can also cause anemia. [S36, 45:34–47:31]

1.8 Example of disorder: benign neoplasms — Nursing problems:

- Benign = noncancerous: abnormal proliferation only, with normal differentiation; malignant = both abnormal. [S37, 47:31–50:34]
- Usually fibrous capsule → contained expansion, no infiltration or metastasis; easier to remove surgically. [S39, 51:10–51:43; sheet p.6]
- Nursing problems are mainly local effects of increasing size/location—not the abnormal substances made by cancer cells.
[S40, 51:43–53:39]
- Can still be fatal if growth interferes with a vital function; benign brain tumors are an example. [S40, 51:43–53:39]

1.9 Interventions (cancer / neoplasms) — blank Interventions column

- Surgery: remove tumor; infiltrating cancer is harder to remove completely, and remaining cells can regrow. [S37, 47:31–50:34; S41, 53:39–54:47]
- Radiation: generates free radicals that damage cancer cells/interfere with replication;

rapidly proliferating cells are most affected, but normal cells can be injured.

[S42, 54:47-55:56; class slide 20]

- Chemotherapy interferes with the cell cycle: S-phase drugs inhibit DNA replication; M-phase drugs prevent division. [S10, 08:59-12:15]
- Different phase targets explain combination drugs and treatment long enough for cells to reach the targeted phase. [S10, 08:59-12:15]
- Chemo effects are not confined to cancer → side effects; cancer treatment may cause anemia. [S43, 55:56-58:23; S36, 45:34-47:31]
- Biologic response modifiers/immunomodulators modify immune responses: monoclonal antibodies and synthetic cytokines. [S44, 58:23-61:14]

1.10 Pathological change: Mutations in regulatory genes — which genes? (proto-oncogenes/oncogenes, tumor-suppressor genes) and PF details

- Genetic mutations alter programming that controls proliferation and differentiation; specific proto-oncogenes/oncogenes and tumor-suppressor genes are not named in these sources. [S13, 15:56-18:11; S14, 18:11-20:45]
- Carcinogens: chemicals, radiation (including sun), cigarette exposure; free-radical DNA damage can cause mutations. Exposure-to-diagnosis lag may be 5-30 years.

[S15, 20:45-23:10; sheet p.6]

- Cancer-causing viruses listed: HPV, EBV, HBV; incorporation of viral DNA can alter host-cell

instructions → abnormal replication.

[sheet p.6; S17, 24:08–25:49]

- Immune defects: AIDS, immunosuppressive drugs, aging → increased cancer incidence. [S18, 25:49–27:10]
- Heredity: germ-cell mutations can pass to children; mutations in regular body cells do not. Cancer risk usually reflects a combination of factors. [S19, 27:10–28:54]

Page 2 — Genetic aspects of the cell

2.1 Gene mutation → Nursing problem:

- DNA replication error/mutation → altered instructions for a protein; radiation or toxic exposure may contribute. [S10, 08:59–12:15; S61, 78:08–79:26]
- Protein is absent or defective/nonfunctional → nursing problems depend on that protein's role. [S62, 79:26–82:02]
- Severity ranges from mild to fatal; no single set of findings applies to all single-gene disorders. [S62, 79:26–82:02]
- Germ-cell mutation can be passed to the next generation. [S61, 78:08–79:26]

2.2 Sex-linked: hemophilia A (factor VIII) — nursing problems/findings

- Hemophilia A: deficiency/nonfunction of factor VIII → assess muscle hematomas and hemarthrosis (joint bleeding). [sheet p.7]
- X-linked recessive pattern: males with a defective X usually have disease; females with

one defective X are usually carriers.

[S66, 84:28-85:47; S67, 85:47-87:03]

- ★ **Prof emphasis:** sex-linked test questions specify a son or daughter—calculate risk using the two possibilities of that sex, not all four children. [S84, 98:48-100:21]

2.3 Autosomal dominant: Marfan's syndrome — nursing problems/findings

- Autosomal dominant: one defective gene causes disease; Aa = disease, aa = normal; no carriers in the taught pattern.
[S72, 88:29-89:57]
- Defective connective tissue → long, thin arms and fingers; musculoskeletal deformities.
[S73, 89:57-90:47; S75, 91:09-91:49]
- Assess eye problems: retinal detachment or lens dislocation from weak connective tissue/ligaments. [S75, 91:09-91:49]
- Assess cardiovascular problems: valve prolapse and aneurysms. [S75, 91:09-91:49]

2.4 Autosomal recessive: sickle cell anemia — nursing problems/findings

- Autosomal recessive: two defective genes needed; aa = disease, Aa = carrier, AA = normal genes. [S76, 91:49-93:44]
- Abnormal RBCs clump together → obstruction/clot-related problems.
[S77, 93:44-95:07]
- Hemolytic component: breakdown of RBCs/hemoglobin contributes to anemia.
[S78, 95:07-95:58]

2.5 Chromosomal disorders: Trisomy 21 (Down syndrome) — findings; also mitosis vs meiosis errors (nondisjunction)

- Mitosis: somatic/body cells; meiosis: germ cells, with 23 chromosomes per egg/sperm → 46 at fertilization. [S4, 01:32-02:10; S9, 07:55-08:59]
- DNA-copying errors can cause gene disorders; chromosome-separation errors cause unequal chromosome numbers. [S10, 08:59-12:15]
- Meiotic nondisjunction: one daughter cell receives both copies, the other none → after fertilization, trisomy or monosomy.
[S95, 117:31-118:28; S96, 118:28-119:10; S98, 120:06-120:38]
- Down syndrome = trisomy 21: three copies of chromosome 21. [S99, 120:38-122:13]
- Findings: characteristic facial features, intellectual disability, possible congenital heart problems and GI malformations.
[S99, 120:38-122:13]
- Structural errors can also occur: a chromosome piece breaks off/is lost → loss of multiple genes. [S93, 114:36-116:33]

2.6 Interventions (genetic disorders) — blank Interventions column

- Single-gene interventions depend on the affected protein and problems caused; no universal treatment. [S62, 79:26-82:02]
- Chromosomal disorders: treat symptoms, maximize quality of life, and help families support development. [S100, 122:13-124:03]

- Genetic counseling: Punnett squares estimate the chance of an affected child, carrier, or child with normal genes. [S80, 96:17–97:13]
- ★ **Prof emphasis:** Punnett squares will be on the test—diagram parents, fill boxes, label each child's status, then calculate the specific risk asked. [S81, 97:13–97:41]
- For her sex-specific X-linked questions, use X/Y and risk out of 2; for autosomal questions, use A/a and risk out of 4.

[S84, 98:48–100:21; S88, 106:29–108:06; S90, 110:58–112:26]